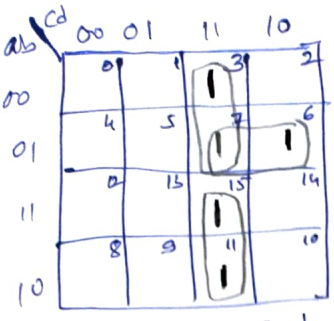
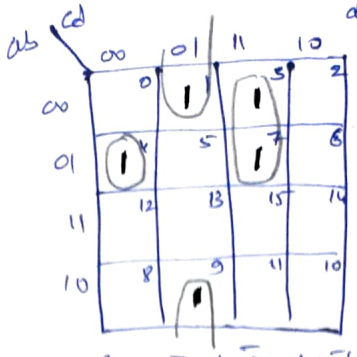
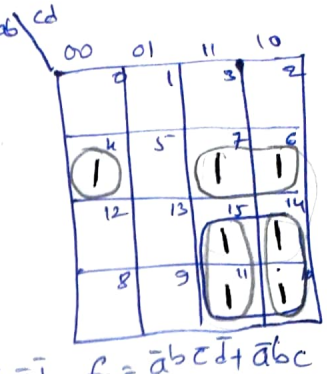
Q-6



$f_1 = \bar{a}cd + \bar{a}bc + ac\bar{d}$



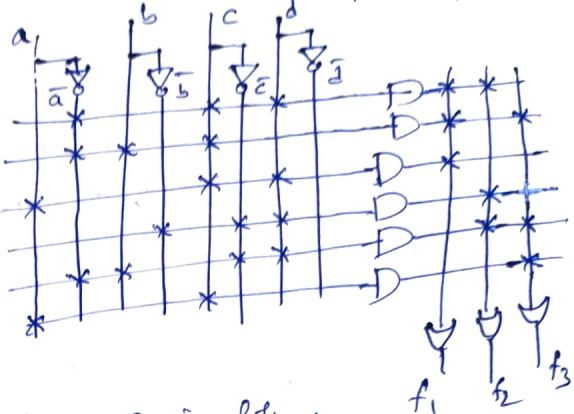
$f_2 = \bar{a}cd + b\bar{c}d + \bar{a}b\bar{c}\bar{d}$



$f_3 = \bar{a}b\bar{c}\bar{d} + \bar{a}bc + ac$

PLA Table

Product	Inputs a	b	c	d	Outputs f_1	f_2	f_3
$\bar{a}cd$	0	–	1	1	1	1	–
$\bar{a}bc$	0	1	1	–	1	–	1
$ac\bar{d}$	1	–	1	0	1	–	–
$b\bar{c}d$	–	0	0	1	–	1	–
$\bar{a}b\bar{c}\bar{d}$	0	1	0	0	–	1	1
ac	1	–	1	–	–	–	1



K-map simplification — 2M

PLA TABLE — 1M

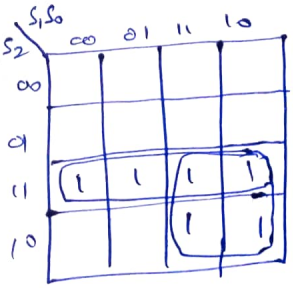
DRAW PLA — 1M

Q-7

Decimal Adjustment Design — 1M

Circuit Diagram — 2M

The output sum should be added with six if input is greater than 9 or carry = 1.



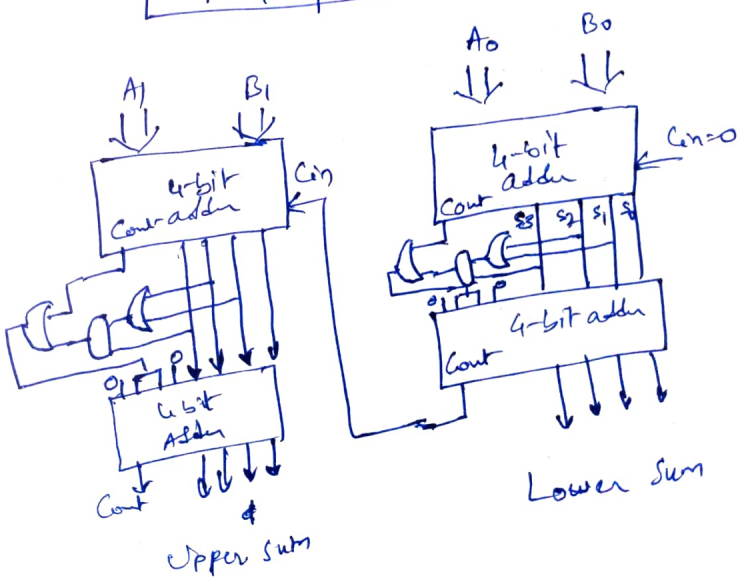
$= S_3S_2 + S_3S_1 = S_3(S_2 + S_1)$

BCD

Let 2 digit Number be

$A_1\ A_0$

$+\ B_1\ B_0$



Q. No. 8

$C_{i+1} = g_i + P_i C_i$. $C_1 = g_0 + P_0 C_0$

$g_0 = x_0 y_0$ $P_0 = (x_0 + y_0)$

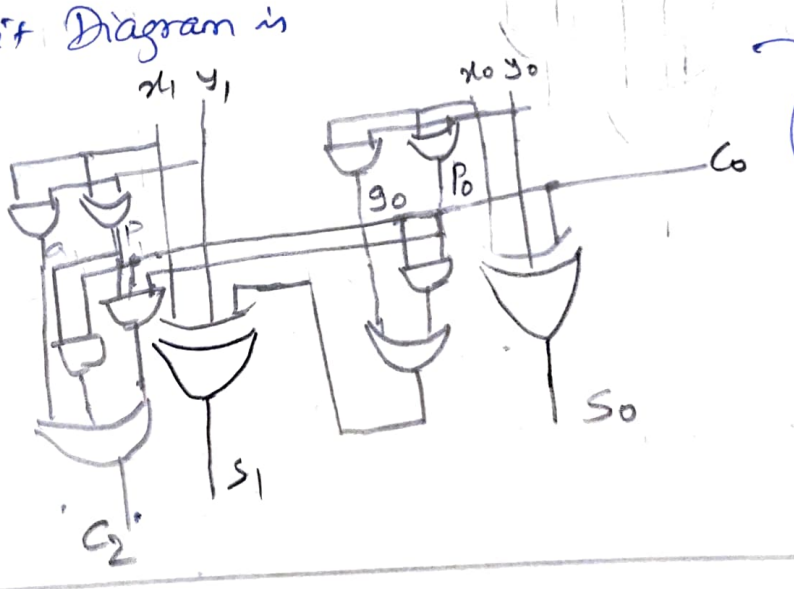
$C_2 = g_1 + P_1 C_1$, $g_1 = x_1 y_1$, $P_1 = (x_1 + y_1)$ } 2M

$\therefore C_2 = g_1 + P_1[g_0 + P_0 C_0] = g_1 + P_1 g_0 + P_1 P_0 C_0$

Expression for sum of least significant Bits $= x_0 \oplus y_0 \oplus C_0 = S_0$

Expression for the sum of most significant bit $S_1 = x_1 \oplus y_1 \oplus C_1$

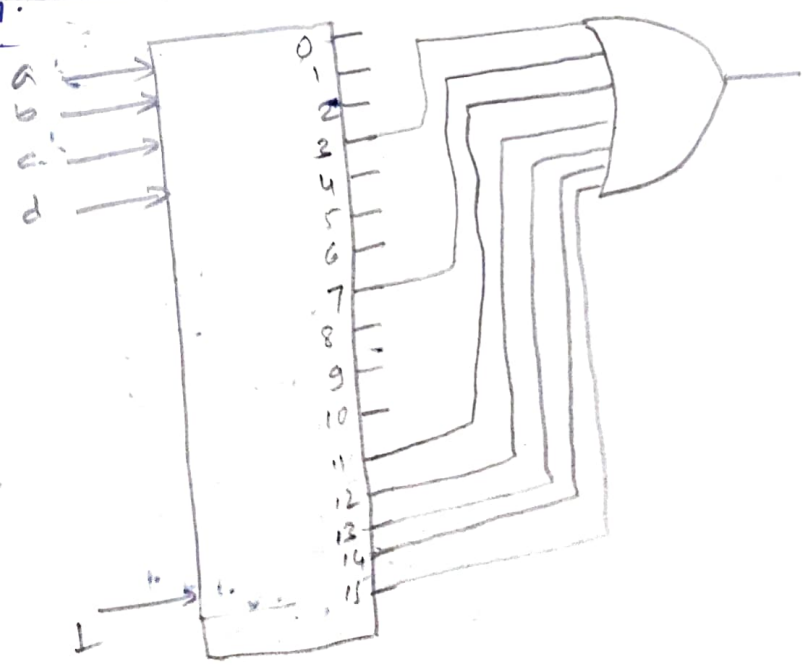
∴ The circuit Diagram is



(1M)

9.

a	b	c	d	f = calculated
0	0	0	0	0
0	0	0	1	0
0	0	1	0	0
0	0	1	1	0
0	1	0	0	0
0	1	0	1	0
0	1	1	0	1
0	1	1	1	0
1	0	0	0	0
1	0	0	1	0
1	0	1	0	0
1	0	1	1	1
1	1	0	0	1
1	1	0	1	1
1	1	1	0	1
1	1	1	1	1

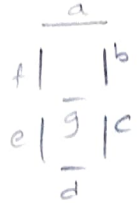


Truth table – 1.5M

Circuit – 1.5M

10

w	x	y	a	b	c	d	e	f	g
0	0	0	1	1	1	1	1	1	0
0	0	1	0	1	1	0	0	0	0
0	1	0	1	1	0	1	1	0	1
0	1	1	1	1	1	1	0	0	1
1	0	0	0	1	1	0	0	1	1
1	0	1	1	0	1	1	0	1	1
1	1	0	1	0	1	1	1	1	1
1	1	1	1	1	1	0	0	0	0



$a = \Sigma 0, 2, 3, 5, 6, 7 = \Pi(1,4)$

$b = \Sigma 0, 1, 2, 3, 4, 7 = \Pi(5,6)$

$c = \Sigma 0, 1, 3, 4, 5, 6, 7 = \Pi(2)$

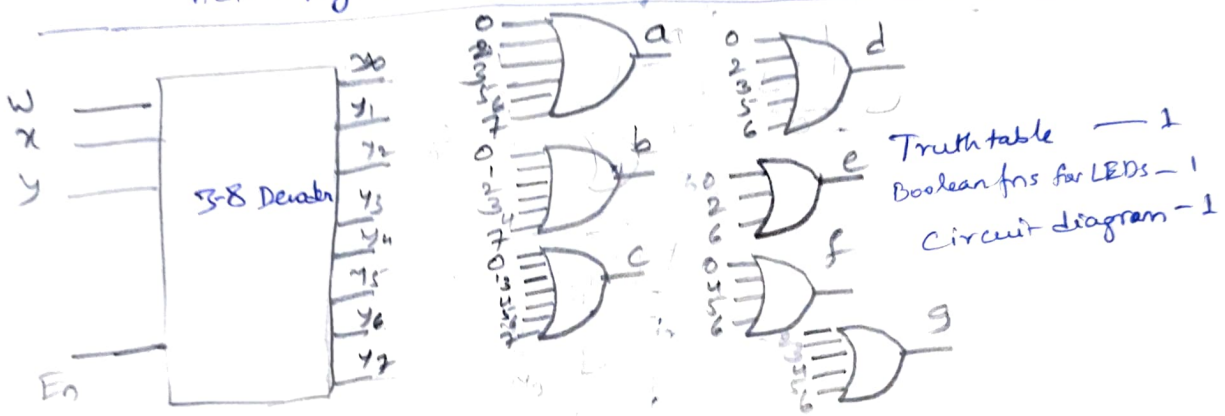
$d = \Sigma 0, 2, 3, 5, 6 = \Pi(1,4,7)$

$e = \Sigma 0, 2, 6 = \Pi(1,3,4,5,7)$

$f = \Sigma 0, 4, 5, 6 = \Pi(1,2,3,7)$

$g = \Sigma 2, 3, 4, 5, 6 = \Pi(0,1,7)$

Appropriate decoder should be 3 to 8 decoder
Active high decoder



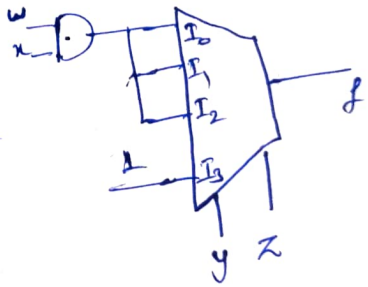
Truth table — 1
Boolean fns for LEDs — 1
Circuit diagram — 1

11

w	x	y	z	f
0	0	0	0	0
0	0	0	1	0
0	0	1	0	0
0	0	1	1	1
0	1	0	0	0
0	1	0	1	0
0	1	1	0	0
0	1	1	1	1
1	0	0	0	0
1	0	0	1	0
1	0	1	0	0
1	0	1	1	1
1	1	0	0	1
1	1	0	1	1
1	1	1	0	1
1	1	1	1	1

Multiplexer table

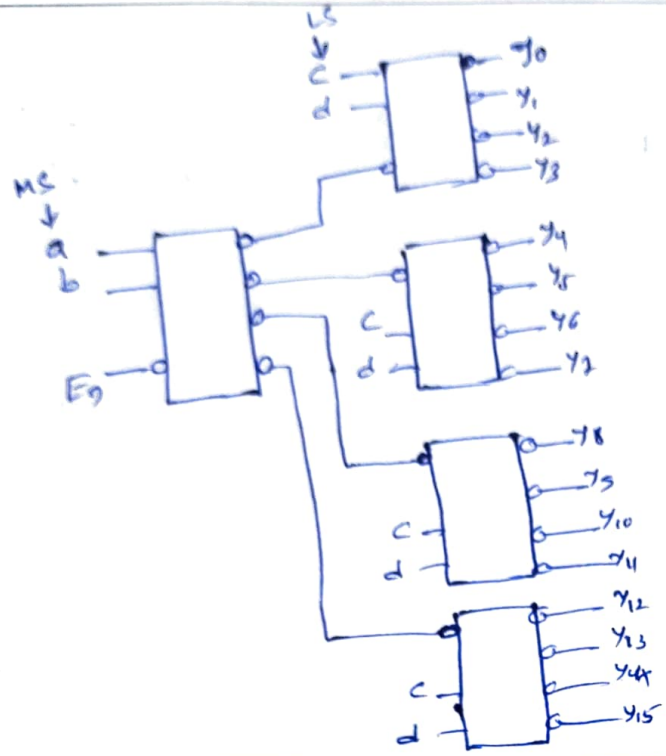
yz \ wx	00	01	10	11	
00	0	0	0	1	→ wx
01	0	0	0	1	→ wx
10	0	0	0	1	→ wx
11	1	1	1	1	→ 1



Truth table — 1
Max table — 1
Circuit — 1

Q. No.

12



(2M)

13

$$f(xyz) = \bar{x}\bar{y}\bar{z} + xz + \bar{y}z + xyz$$

$$= \bar{x}\bar{y}\bar{z} + \bar{y}z + xz(1+y)$$

$$= \bar{x}\bar{y}\bar{z} + \bar{y}z + xz$$

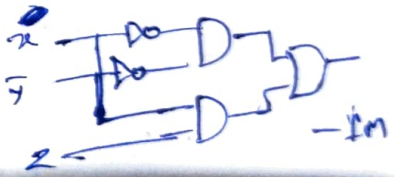
$$= \bar{y}[z + \bar{z}\bar{x}] + xz$$

$$= \bar{y}[z + \bar{x}] + xz \quad \because \text{consensus}$$

$$= \bar{y}z + \bar{x}\bar{y} + xz$$

$$= \bar{x}\bar{y} + xz$$

— (1M)



— 1M

14 —

$$F(X,Y,Z) = (X+Y)(\bar{Y}+XZ)(X+\bar{Z})$$

$$\text{CPOS} = (X+Y+Z)(X+Y+\bar{Z})(X+\bar{Y}+Z)(X+\bar{Y}+\bar{Z})(\bar{X}+\bar{Y}+Z) \quad \text{—(1M)}$$

$$\text{CSPP} = XYZ + X\bar{Y}Z + X\bar{Y}\bar{Z} \quad \text{—(1M)}$$